BACSE344 - Cloud Infrastructure and Architecture

Digital Assignment 3 Question

C2 Slot

Last date to submit in MSTeams : 30/10/2026

General instruction:

Continue with the same application you built and deployed in Digital Assignment 1 (hospital, institute, industry or any other domain). Every answer must be based on your own application and domain.

Question 1 - Scale your application (5 marks)

Your application's user base is about to grow sharply. Identify a realistic **peak-demand situation for your own domain** (for example, an outpatient rush in a hospital, results day at an institute, or a production shift change in a factory) and make your application handle it without downtime.

a) Describe your peak-demand situation and the API that will be hit hardest. Then run **three instances** of your backend using Docker Compose behind an **Nginx load balancer**, connected to your existing cloud database. Show that requests are distributed across all three instances. **(1 mark)**

b) Using **Apache Bench or any suitable free load-testing tool** (for example, hey, JMeter or k6), send 1,000 requests to that API, first with 1 instance and then with 3 instances, **keeping the same tool and concurrency level for both runs**. Tabulate requests per second and average response time, and state the improvement. **(1.5 marks)**

c) Stop one instance while the load test is running and record the failed requests. Then configure Nginx so that **no request fails**, and show the before-and-after count. **(1.5 marks)**

d) Draw the architecture diagram of **your own implementation**, using the **actual names** of your components: your frontend, the three backend containers and Nginx from part (a) (with their ports), your cloud database and hosting platform from Assignment 1, and the tools used (e.g., MongoDB Atlas, Render, Firebase). Then add the **cloud bursting** or **edge/fog computing** component to the same diagram, marked in a different colour, and explain how it would handle **your peak-demand situation** from part (a), and why you chose it for your domain. **(1 mark)**

Question 2 - Secure your application (5 marks)

Your application stores data about real people in your chosen domain. Secure it before it goes live.

a) Identify the **most sensitive data** your application stores (for example, patient records, student marks, or employee/production data). Store user passwords as **salted hashes** (bcrypt or Argon2) instead of plain text, and show a database screenshot before and after. **(1 mark)**

b) Add **multi-factor authentication** to login using a TOTP authenticator app (Google or Microsoft Authenticator). Restrict your **Update and Delete APIs** to logged-in users with the right role **for your domain** (for example, doctor, faculty, or supervisor), and show a request being rejected without a valid token. **(1.5 marks)**

c) Scan your application's Docker image with **Trivy**. **Harden** it (smaller base image, non-root user), rescan, and tabulate the CRITICAL and HIGH counts before and after. **(1.5 marks)**

d) Name **two threat agents** relevant to your domain. For each, state the control you implemented against it and one requirement from the **regulation that applies to your**

application (HIPAA for health data, GDPR for personal data in general), with a reason why that regulation applies. **(1 mark)**

What to submit

- A report with screenshots, result tables for every part, and the GitHub link to an assignment-3 branch of your assignment 1 repository.
- A demo video of 3 minutes.

Evaluation rubric

The same scale applies to every part of both questions.

Evidence	Marks
Working, with screenshots/tables and a clear explanation	Full marks for that part
Works, but evidence or explanation is missing or generic	Half marks
Not attempted or not working	Zero